Context Free Grammer (CFG)

$G = (V, T, P, S)$

Q. Construct context free grammar which accept the string having atleast 2a's over $\Sigma = \{a, b\}$

$RE = (a+b)^* a (a+b)^* a (a+b)^*$

$S = TaTaT$

$T \rightarrow aT \mid bT \mid \epsilon$ — Production

$V = \{S, T\}$

$T = \{a, b, \epsilon\}$

S : Start symbol.

i) aba

LMD

$S \rightarrow TaTaT$
$\rightarrow \epsilon aTaT$
$\rightarrow aTaT$
$\rightarrow abTaT$
$\rightarrow ab\epsilon aT$
$\rightarrow abaT$
$\rightarrow aba\epsilon$
$\rightarrow aba$

$S \rightarrow TaTaT$
$\rightarrow TaTa\epsilon$
$\rightarrow TaTa$
$\rightarrow TabTa$
$\rightarrow Tab\epsilon a$
$\rightarrow Taba$
$\rightarrow \epsilon aba$
$\rightarrow aba$

Q. $\Sigma = \{a, b\}$

$E \rightarrow E+E$

$E \rightarrow E*E$

$E \rightarrow a/b/c$

$a+b*c$

$V = E$

$T = \{a, b, c, +, *\}$

$S = \{E\}$

$E \rightarrow E+E/E*E/a/b/c$

i) $E \rightarrow E+E$ $E \rightarrow a$ LMD

$\rightarrow a+E$ $E \rightarrow E+E$

$\rightarrow a+E*E$ $E \rightarrow b$

$\Rightarrow a+b*E$ $E \rightarrow c$

$\rightarrow a+b*c$

$\Rightarrow a+b*c$

ii) $E \rightarrow E*E$ LMD $E \rightarrow E+E$

$\rightarrow E+E*E$

$\rightarrow a+E*E$

$\rightarrow a+b*E$

$\rightarrow a+b*c$

iii) RMD

$E \rightarrow E+E$

$E \rightarrow E+E*E$

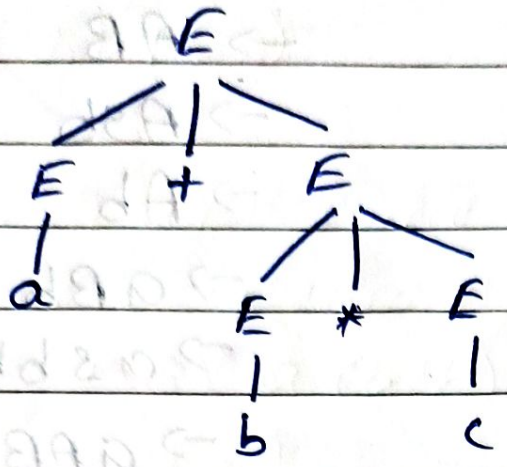
$E \rightarrow E+E*c$

$E \rightarrow E+b*c$

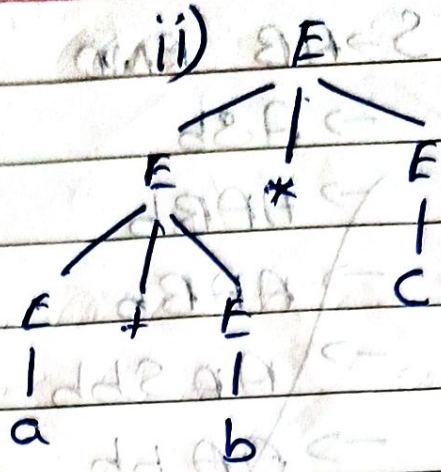
$E \rightarrow a+b*c$

$\Rightarrow$

i) Parse tree



ii)



If there is 2LMD & 2RMD then its called ambiguous.

$S \to AB/\epsilon$

$A \to aB$ aabbbb

$B \to Sb$

~~$S \Rightarrow AB$ $A \to aB$~~

$S \to AB$ LMD

$\to aBB$

$\to aSbB$

$\to aABbB$

$\to aaBBbB$

$\to aaSbBbB$

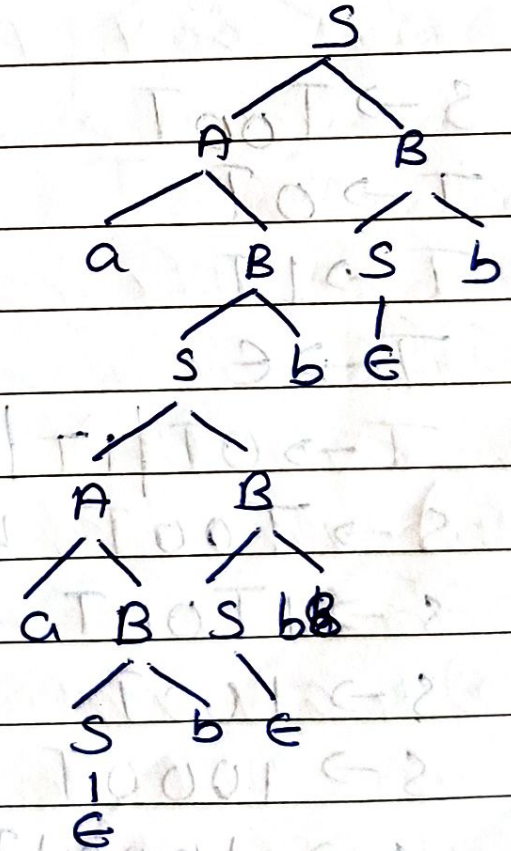
$\to aaSbSbbB$

$\to aa\epsilon b\epsilon bbSb$

$\to aabbb\epsilon b$

$\to aabbbb$

$=$



It is not ambiguous/ unambiguous.

S → AB RMD
→ ASb
→ AABb
→ AABb
→ AASbb
→ AAbb
→ AaABbb
→ AaSbbb
→ Aabbbb
→ aBabbb
→ asbabbb
→ ababbb

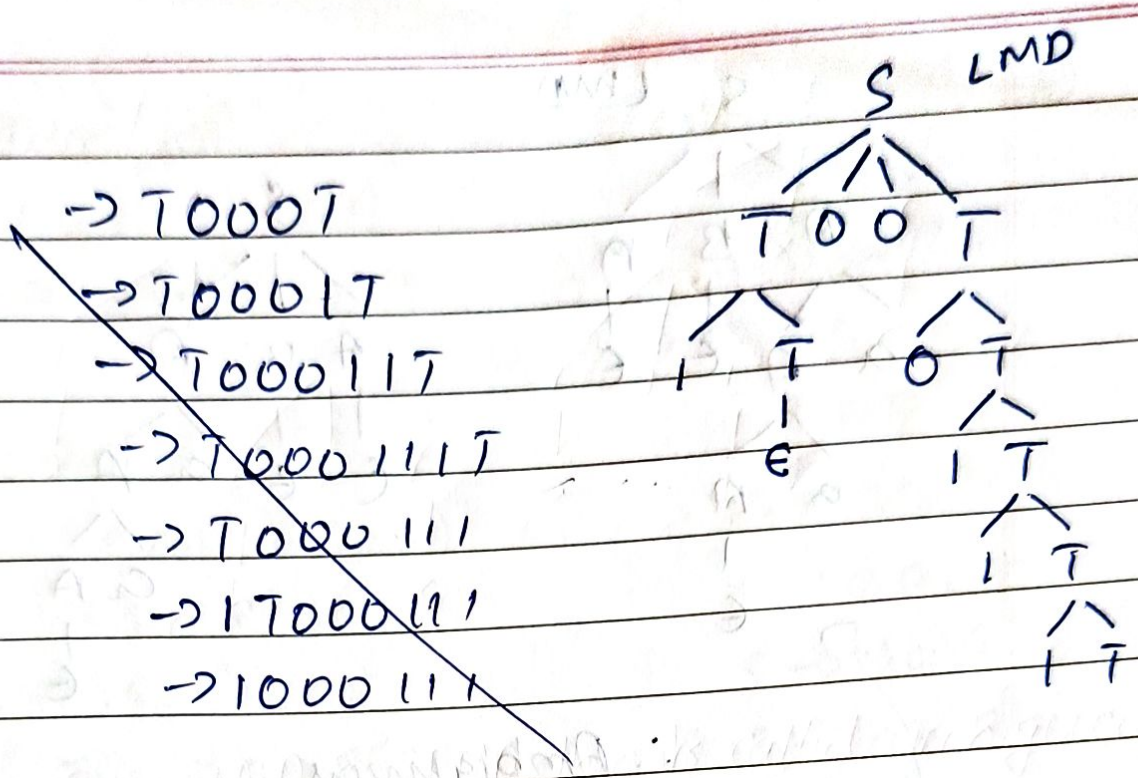
→ AB
→ ASb
→ Ab
→ aBb
→ asbb
→ aABbb
→ aASbbb
→ aAbbb
→ aaBbbb
→ aaSbbbb
→ aabbbb

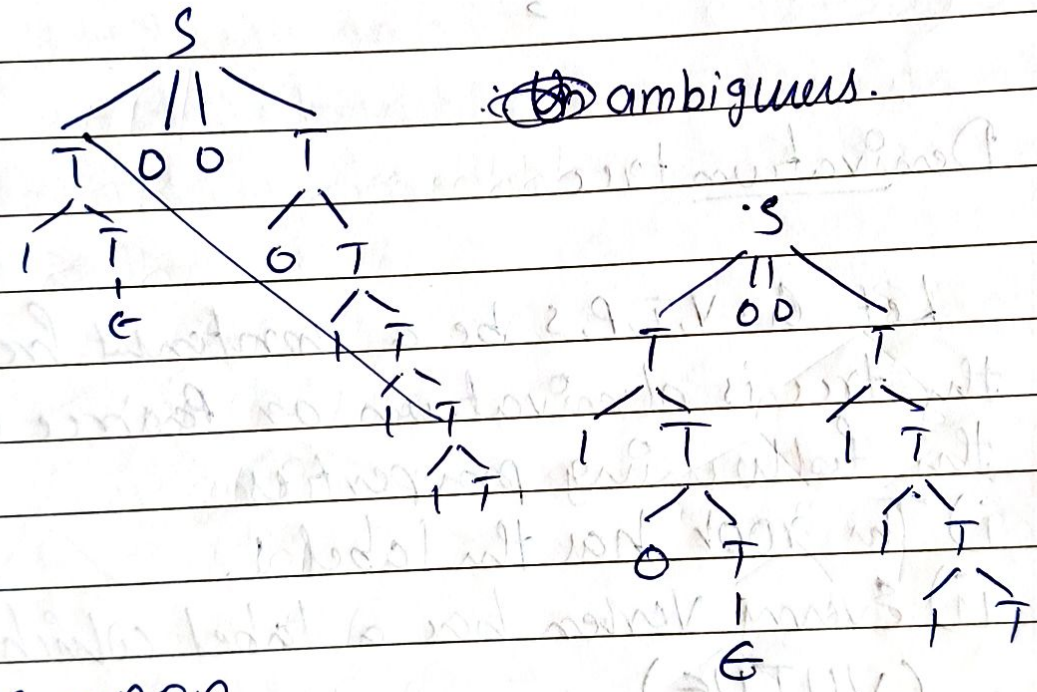
Q. S → TooT
T → oT
T → 1T
T → ∈
T → oT | 1T | ∈
S → TooT LBMD
S → 1TooT
S → 10oT
S → 100T
S → 1000T
S → 10001T
S → 100011T
S → 1000111T
S → 1000111

1000111

S → TOOT RMD
→ TOOIT → TOO1T
→ TOO1IT → TOO11
→ TOO111T → 10TOO111
→ TOO111 → 10TOO111
→ T0 → 100111

classmate
Date
Page





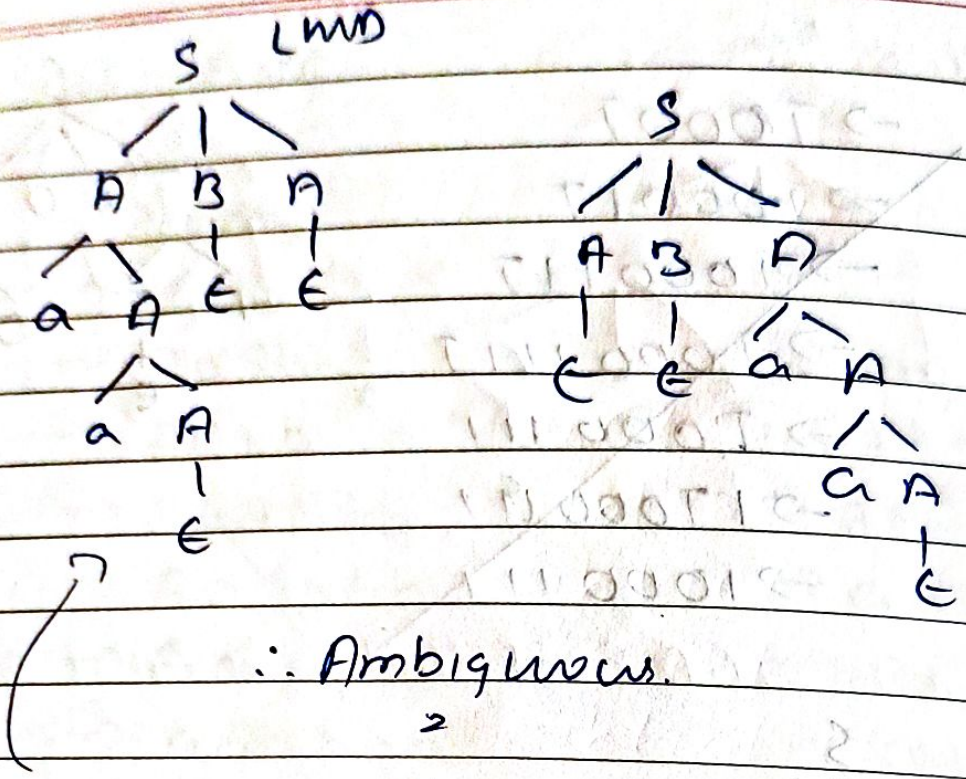
∴ ambiguous.

Q. S → ABA
A → aA | ε
B → bB | ε

aa

S → ABA
→ aABA
→ aaABA
→ aaεBA
→ aaBA
→ aaεA
→ aaε

S → ABA
→ ABaA
→ ABaaA
→ ABaaε
→ Aεaa
→ εaa
→ aa



∴ Ambiguous.

Derivation tree

Let $G = V, T, P, S$ be a context free grammar then tree is derivation or Parse tree with the following properties.

i) The root has the label S.

ii) Every vertex has a label which is in $(V \cup T \cup \epsilon)$

iii) Every leaf node has label from $T \cup \epsilon$ an interior vertex has a label from V

iv) If a vertex is labeled A and if $x_1, x_2 \ldots x_n$ are all children of A from left then $A \to x_1, x_2, x_3 \ldots x_n$ must be a production in P.

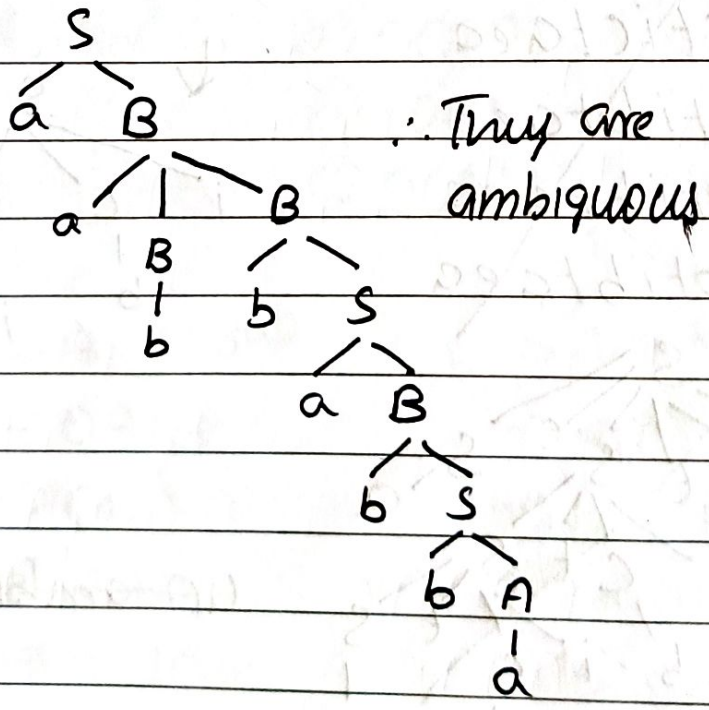
Q. $S \to aB \mid bA$

$A \to a \mid aS \mid bAA$

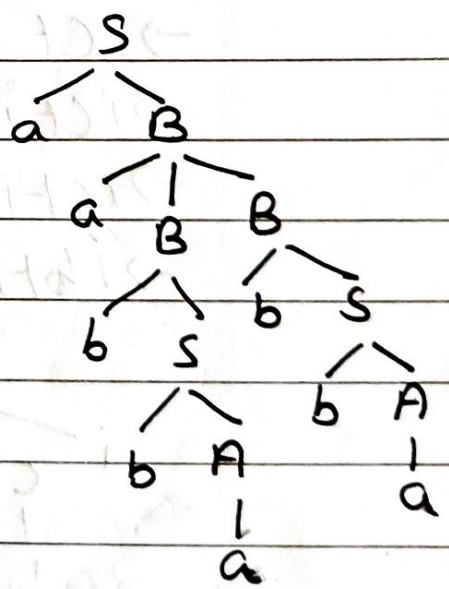
$B \to b \mid bS \mid aBB$ aabbabba

$S \to aB$ (RMD)
$\to aaBB$
$\to aabB$
$\to aabbS$
$\to aabbaB$
$\to aabbabS$
$\to aabbabbA$
$\to aabbabba$

$S \to aB$
$S \to aaBB$
$S \to aaBbS$
$S \to aaBbbA$
$S \to aaBbba$
$S \to aabsbba$
$S \to aabbAbba$
$S \to aabbabba$



∴ They are ambiguous



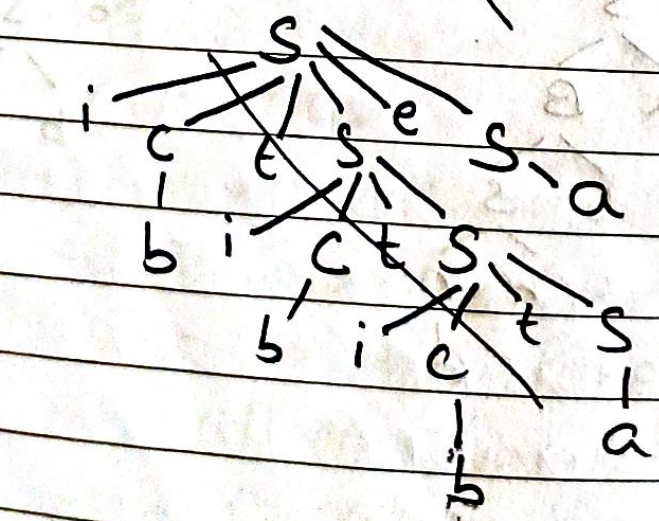
Q. Check whether the given grammar is ambiguous or not.

ibtibtibtaea

S → iCts
S → iCtSeS
S → a
C → b

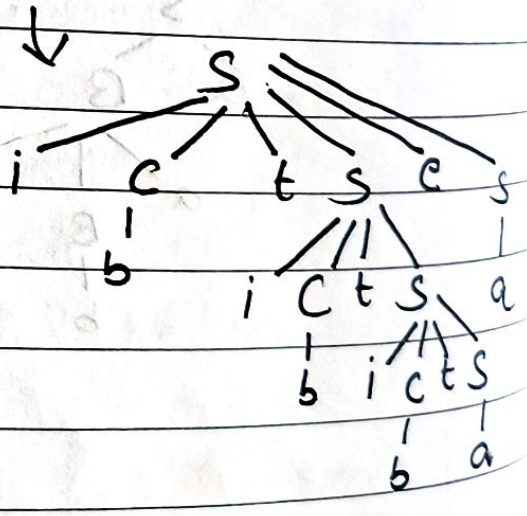
RDM

S → iCtSeS
→ iCtsea
→ iCtiCtSea
→ iCtibtsea
→ iCtiCtiCtsea
→ iCtiCtiCtaea
→ iCtiCtibtaea
→ iCtibtibtaea
→ ibtibtibtaea



LDM

S → iCtSeS
→ ibtSeS
→ ibtiCtSeS
→ ibtibtSeS
→ ibtibtiCtSeS
→ ibtibtibtSeS
→ ibtibtibtaeS
→ ibtibtibtaea



~~Un-ambiguous~~

RDM

$S \to iCts$
$\to ictiCts$
$\to ictictictses$
$\to ictictictaea$
$\to ictictibtaea$
$\to ictibtibtaea$
$\to ibtibtibtaea$
$=$

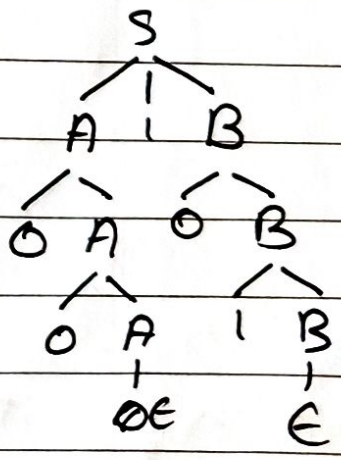
2. $S \to AB \mid AIB$
$A \to OA \mid \epsilon$
$B \to OB \mid IB \mid \epsilon$

a) 00101
b) 1001
c) 00011

a) 00101

$S \to AIB$ LMD
$\to OAIB$
$\to OOAIB$
$\to OOIB$
$\to OOIOB$
$\to OOIOI$

$S \to AIB$ RMD
$\to AIOB$
$\to AIOI$
$\to OAIOI$
$\to OOIOI$



unambiguous.

b) 1001

$S \to A|B$ LMD

$\to 1B$

$\to 10B$

$\to 100B$

$\to 1001B$

$\to 1001$

$S \to A|B$

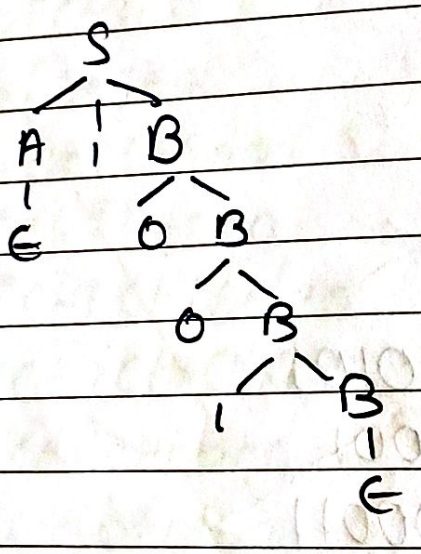
$S \to A1$

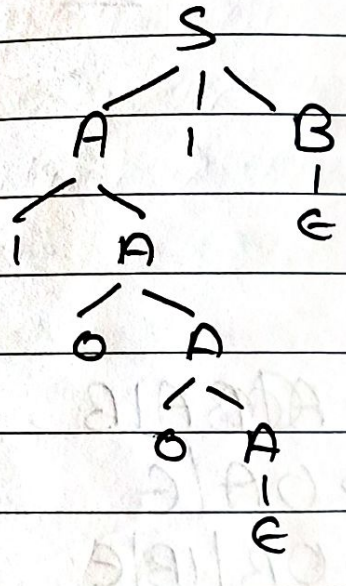
$S \to 1A1$

$S \to 10A1$

$S \to 100A1$

$S \to 1001$





ambiguous

c) 00011

$S \to A|B$

$\to 0A1B$

$\to 0001B$

$\to 00011B$

$\to 00011$

$S \to A|B$

$\to A11B$

$\to A11\varepsilon$

$\to 00011$

unambiguous

Removal of ambiguity

1) Precedance
2) Associativity.

Q id + id * ic

$E \to E+T \mid T$
$T \to T*F \mid F$
$F \to id$

$E \to E+T$
$\to T+T$
$\to F+T$
$\to id+T$
$\to id+T*F$
$\to id+F*F$
$\to id+id*id$

∴ The given grammar is unambiguous

=

Q. $E \to E+E \mid E-E \mid E*E \mid E/E \mid (E) \mid id \mid E^{\wedge}E$

id ()
* /
\+ -
} LEFT

$E \to E+T \mid E-T \mid T$
$T \to T-S \mid S$
$S \to S*U \mid U$
$U \to U \mid V \mid V$
$V \to$

$T \to T*F \mid T/F \mid F$
$F \to (E) \mid id$
id + (id * id)

$E \to E+T \to T+T \to F+T \to id+T$
$\to id+F \to id+(E) \to id+(T) \to id+(T*F)$
$\to id+(id*id)$

=

Q. $E \to E+E / E-E / E*E / E/E / (E) / id / E^{\wedge}E$

$E \to E+T / E-T / T$
$T \to T*F / E/T / F$
$F \to P \wedge F / P$
$P \to (E) / id$

Q. $E \to E+E / E*E / id$
construct a un ambiguous grammer where + has higher precidence than you combine with the star & + is right associative.

$E \to E*T / T$
$T \to F+T / F$
$F \to id$

$id + id * id$

$T \to F+T$
$T \to id+T$

$E \to E*T$
$\to T*T$
$\to F+T*T$
$\to id+T*T$
$\to id+F*F$
$\to id+id*id$

Q. To write the context free grammar for.
L = set of all palindromes over a,b.

$S \to \epsilon$

$S \to a$ } a/b

$S \to b$ }

$S \to aSa \mid bSb$ abba

$S \to aSa$

$S \to abSba$

$S \to abba$.

Q. $L = \{a^n b^n : n \geq 0\}$

$S \to \epsilon$

$S \to ab$ asb

$\Rightarrow S \to \epsilon / asb$

aaabbb

$S \to asb$

$S \to aasbb$

$S \to aaasbbb$

$S \to aaabbb$

Q. $L = \{a^n b^n : n \geq 1\}$

$S = aWb$

$W = \epsilon / S$

aaa bbb

$S \to aWb$

$\to asb$

$\to aawbb$

$\to aasbb$

$\to aaawbbb$

$\to aaabbb$

V7 HIGH-TECHPOINT 0.7

Q $L = \{a^{n+1} b^n : n \geq 0\}$

$S \rightarrow aAb$ $S \rightarrow a/aSb$
$A \rightarrow \in/a$

Q $L = \{a^n b^{n+1} : n \geq 0\}$
$S \rightarrow b/aSb$

Q $L = \{a^n b^{n+2} : n \geq 0\}$
$S \rightarrow abb/aSb$

Q $L = \{a^n b^{2n} : n \geq 0\}$

$S \rightarrow \in/aAb$
$A \rightarrow$ $S \rightarrow abb/aSbb$

abb
aabbb
aabb

Q $L = \{0^m 1^m 2^n : m \geq 1 \& n \geq 0\}$
A B
$0 \leq 1$
$S \rightarrow 01/0S1/A2/0S1$
$A \rightarrow \in/A2$

$S \rightarrow AB$
$A \rightarrow 01/0A1$
$B \rightarrow \in/2B$

$A = 0^m 1^m$
$= \{01, 0011$
$A \rightarrow 01/0A1$
$B \rightarrow \in/2B$

Q. W starts and ends with the same symbol
$\Sigma = \{a, b\}$

a w a
b w b

ababa

$S \to a/b/awa/bwb$
$w \to awb/bwa/\epsilon/abwa/wb/awa/bwa$

e.g. $S \to awa$

abab bbbba

awa
abwba
abawbba
ababwbbba
ababεbbba

Or

$S \to aAa/bAb/a/b/\epsilon$
$A \to aA/bA/\epsilon$

Q $L = \{a^i b^j c^k : j = i + k \quad i \geq 0, k \geq 0\}$

$A \to aA/\epsilon$ aAb / ε
$B \to bC/\epsilon$ bCc / ε
$S \to AB$

Q. $L = \{w / |w| \text{ is odd}\}$

$A \to aAbb / aaAb$

$S \to a / b / A$

$w \to a / b / awa / bwb / awb / bwa$

Q. $L = \{w / w = w^R\}$ i.e. w is palindrome

$w \to awa / bwb / a / b / \in$

Q. $L = \{a^i b^j c^k / i = j + k\}$

$A \to aAb / \in$

$B \to aBc / \in$

$S \to A / B$

$a^5 b^3 c^2$

$aABc$

$aaABBc$

a^2ABc^2

a^3ABbc^2

a^3ABbc^2

$aABc$

$aaAbBc$

$aaAbaABc$

$S \to aSc / A$

$A \to aAb / \in$

Build Your Dream classmate Date Page

PDA

$L = \{a^n b^n : n \geq 1\}$

aabb ⇒

a
a
Z_0

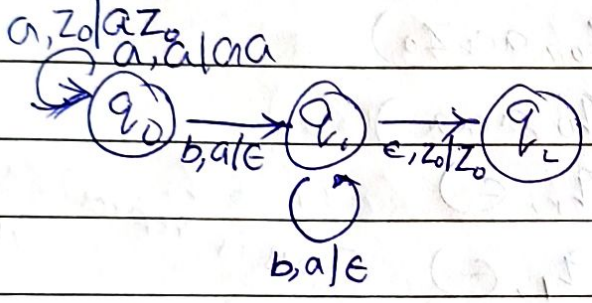
$\delta(q_0, a, Z_0) = (q_0, aZ_0)$

$\delta(q_0, a, a) = (q_0, aa)$

$\delta(q_0, b, a) = (q_1, \epsilon)$

$\delta(q_1, b, a) = (q_1, \epsilon)$

$\delta(q_1, \epsilon, Z_0) = (q_2, Z_0)$ or (q_2, ϵ)



$Q = \{q_0, q_1, q_2\}$

$\Sigma = \{a, b\}$

$\Gamma = \{a, Z_0\}$

$q_0 = q_0$.

Q. Write instantaneous description for the string aaabbb

$(q_0, aaabbb, Z_0) \vdash (q_0, aabbb, aZ_0)$

$\vdash (q_0, abbb, aaZ_0)$

$\vdash (q_0, bbb, aaaZ_0)$

$\vdash (q_1, bb, aaaZ_0)$

$\vdash (q_1, b, aaZ_0)$ Valid

$\vdash (q_1, \epsilon, aZ_0) \vdash (q_2, \epsilon, Z_0)$

$\vdash (q_2, \epsilon, \epsilon)$

abba

$(q_0, abba, z_0) \vdash (q_0, bba, az_0)$

$\vdash (q_1, ba, z_0)$

Q $L = \{a^n b^{2n} / n \geq 1\}$

$L = \{abb, aabbbb\}$

As b is 2 times a while considering n as 1 push 2 a's



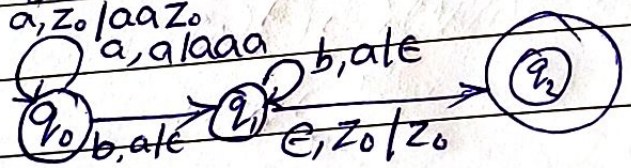
$\delta(q_0, a, z_0) = (q_0, aaz_0)$

$\delta(q_0, a, a) = (q_0, aaa)$

$\delta(q_0, b, a) = (q_1, \epsilon)$

$\delta(q_1, b, a) = (q_1, \epsilon)$ This is repeating so stop as it's a loop.

$\delta(q_1, \epsilon, z_0) = (q_2, z_0)$



$(q_0, aaabbbbbb, z_0) \vdash (q_0, aabbbbbb, aaz_0)$

$\vdash (q_0, abbbbbb, aaa)$

$\vdash (q_0, bbbbbb, aaa)$

$\vdash (q_0, bbbbb, aaa)$

$\vdash (q_{01}, bbbb, aaa)$

$\vdash (q_{01}, bbb, aaa)$

$\vdash (q_1, bb, aaz_0)$

$\vdash (q_1, b, az_0)$

$\vdash (q_1, \epsilon, z_0) \vdash (q_2, \epsilon, z_0)$

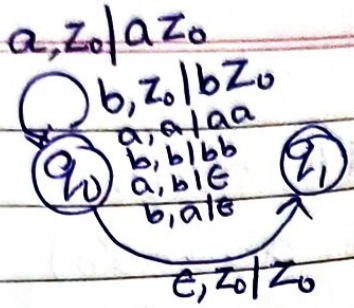
classmate
Date 23/10/26
Page

aab
$(q_0, aab, z_0) \vdash (q_0, ab, aaz_0)$
$\vdash (q_0, b, aaaaz_0)$
$\vdash (q_1, \epsilon, aaaz_0)$

abb
$(q_0, abb, z_0) \vdash (q_0, bb, aaz_0)$
$\vdash (q_1, b, az_0)$
$\vdash (q_1, \epsilon, z_0) \vdash (q_2, \epsilon, z_0)$

Equal no. of a's & b's
{ab, abab, ...}
$(q_0, abab, z_0) \vdash (q_0, bab, aaz_0)$
$\vdash (q_0, bab, az_0)$
$\vdash (q_0, b, aaaz_0)$
$\vdash (q_0, \epsilon, aaz_0)$ ∝

Q. Obtain a PDA for:
$L(M) = \{w / w \in (a+b)^* \text{ and } n_a(w) = n_b(w)\}$
$L = \{ab, abba, abab, baba\}$
$\delta(q_0, a, z_0) = (q_0, az_0)$
$\delta(q_0, b, a) = (q_0, \epsilon)$
$\delta(q_0, \epsilon, z_0) = (q_1, z_0)$



→ Instantaneous for abaabb

$(q_0, abaabb, z_0) \vdash (q_0, baabb, az_0)$
$\vdash (q_0, aabb, z_0)$
$\vdash (q_0, abb, az_0)$
$\vdash (q_0, bb, aa)$
$\vdash (q_0, b, az_0)$
$\vdash (q_0, \epsilon, z_0) \vdash (q_1, \epsilon, z_0)$

Construct PDA $L(M) = \{ w c w^R : w \in (a+b)^* \}$
where w^R is reverse by a final state.
$L = \{abbcbba\}$
↓ Change state.

$\delta(q_0, a, z_0) = (q_0, az_0)$
$\delta(q_0, a, a) = (q_0, aa)$
$\delta(q_0, b, a) = (q_0, ba)$
$\delta(q_0, b, b) = (q_0, bb)$
$\delta(q_0, c, b) = (q_1, bb)$
$\delta(q_1, b, b) = (q_1, \epsilon)$
$\delta(q_1, b, b) = (q_1, \epsilon)$
$\delta(q_1, a, a) = (q_1, \epsilon)$

$\delta(q, a, a) = (q, \epsilon)$

$\delta(q_0, a, Z_0) = (q_0, aZ_0)$

$\delta(q_0, b, Z_0) = (q_0, bZ_0)$

$\delta(q_0, a, a) = (q_0, aa)$

$\delta(q_0, b, b) = (q_0, bb)$

$\delta(q_0, a, b) = (q_0, ab)$

$\delta(q_0, b, a) = (q_0, ba)$

$\delta(q_0, c, Z_0) = (q_1, Z_0)$

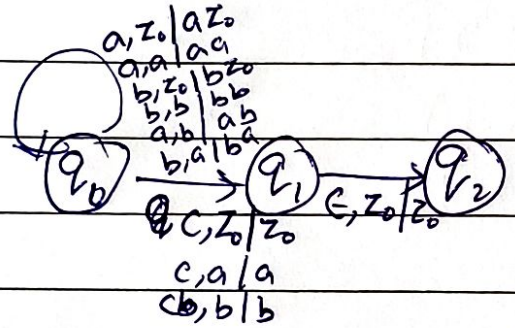
$\delta(q_0, c, a) = (q_1, a)$

$\delta(q_0, c, b) = (q_1, b)$

$\delta(q_1, a, a) = (q_1, \epsilon)$

$\delta(q_1, b, b) = (q_1, \epsilon)$

$\delta(q_1, \epsilon, Z_0) = (q_2, \epsilon Z_0)$



abcba

$(q_0, abcba, Z_0) \vdash (q_0, bcba, aZ_0)$

$\vdash (q_0, cba, baZ_0)$

$\vdash (q_1, ba, baZ_0)$

$\vdash (q_1, a, aZ_0)$

$\vdash (q_1, \epsilon, Z_0) \vdash (q_2, \epsilon, Z_0)$